

# Multiplying Polynomials in Python

At this point, you have created a nice toolbox of functions for dealing with lists of coefficients as polynomials. Create a file called `poly.py` and copy the following functions into it:

- `evaluate_polynomial`
- `polynomial_to_string`
- `add_polynomials`
- `scalar_polynomial_multiply`
- `subtract_polynomial`

Now, create another file in the same directory called `test.py`. Type this into that file:

```
1  import poly
2
3  polynomial_a = [9.0, -4.0, 3.0, -5.0]
4  print('Polynomial A =', poly.polynomial_to_string(polynomial_a))
5
6  polynomial_b = [-9.0, 0.0, 4.0, 2.0, 1.0]
7  print('Polynomial B =', poly.polynomial_to_string(polynomial_b))
8
9  # Evaluation
10 value_of_b = poly.evaluate_polynomial(polynomial_b, 3)
11 print('Polynomial B at 3 =', value_of_b)
12
13 # Adding
14 a_plus_b = poly.add_polynomials(polynomial_a, polynomial_b)
15 print('A + B =', poly.polynomial_to_string(a_plus_b))
16
17 # Scalar multiplication
18 b_scalar = poly.scalar_polynomial_multiply(-3.2, polynomial_b)
19 print('-3.2 * Polynomial B =', poly.polynomial_to_string(b_scalar))
20
21 # Subtraction
22 a_minus_b = poly.subtract_polynomial(polynomial_a, polynomial_b)
23 print('A - B =', poly.polynomial_to_string(a_minus_b))
```

When you run it, you should get the following:

```
1 Polynomial A = -5.0x^3 + 3.0x^2 + -4.0x + 9.0
2 Polynomial B = 1.0x^4 + 2.0x^3 + 4.0x^2 + -9.0
3 Polynomial B at 3 = 162.0
4 A + B = 1.0x^4 + -3.0x^3 + 7.0x^2 + -4.0x
5 -3.2 * Polynomial B = -3.2x^4 + -6.4x^3 + -12.8x^2 + 28.8
6 A - B = -1.0x^4 + -7.0x^3 + -1.0x^2 + -4.0x + 18.0
```

You are now ready to implement the multiplication of polynomials. The function will look like this:

```
1 def multiply_polynomials(a, b):
2     ...Your code here...
```

It will return a list of coefficients.

In an exercise in the last chapter, you were asked “ Let’s say I have two polynomials,  $p_1$  and  $p_2$ .  $p_1$  has degree 23.  $p_2$  has degree 12. What is the degree of their product?” The answer was  $23 + 12 = 35$ .

In our implementation, a polynomial of degree 23 is held in a list of length 24.

In Python, we will be trying to multiply a polynomial  $a$  and a polynomial  $b$  represented as lists. What is the degree of that product?

```
1 result_degree = (len(a) - 1) + (len(b) - 1)
```

Now, we need to create an array of zeros that is one longer than that. Here is a cute Python trick: If you have a list, you can replicate it using the `*` operator.

```
1 a = [5,7]
2 b = a * 4
3 print(b)
4 # [5, 7, 5, 7, 5, 7, 5, 7]
```

Here is how you will get a list of zeros:

```
1 result = [0.0] * (result_degree + 1)
```

We will step through  $a$ , getting the index and value of each entry. You can do this in one

line using enumerate:

```
1 for a_degree, a_coefficient in enumerate(a):
```

For each of those, we will step through the entire b polynomial. As you multiply together each term, you will add it to the appropriate coefficient of the result.

Here is the whole function:

```
1 def multiply_polynomials(a, b): # What is the degree of the resulting  
2 polynomial? result_degree = (len(a) - 1) + (len(b) - 1)  
3  
4     # Make a list of zeros to hold the coefficients result = [0.0] * (result_degree + 1)  
5     (result_degree + 1)  
6  
7     # Iterate over the indices and values of a for a_degree,  
8     a_coefficient in enumerate(a):  
9  
10        # Iterate over the indices and values of b for b_degree,  
11        b_coefficient in enumerate(b):  
12  
13            # Calculate the resulting monomial coefficient =  
14            a_coefficient * b_coefficient degree = a_degree + b_degree  
15  
16            # Add it to the right bucket  
17            result[degree] = result[degree] + coefficient  
18  
19     return result
```

Take a long look at that function. When you understand it, type it into `poly.py`.

In `test.py`, try out the new function:

```
1 # Multiplication  
2 a_times_b = poly.multiply_polynomials(poly.polynomial_a, polynomial_b)  
3 print('A x B =', poly.polynomial_to_string(a_times_b))
```

This is an example of a *nested loop*. The outer loop steps through the polynomial a. For each step it takes, the inner loop steps through the entire polynomial b.

## 1.1 Something surprising about lists

You can imagine that you might want to create two very similar polynomials. Let's say polynomial c is  $x^2 + 2x + 1$  and polynomial d is  $x^2 - 2x + 1$ . You might think you are very

clever to just alter that degree 1 coefficient like this:

```
1 c = [1.0, 2.0, 1.0]
2 d = c
3 d[1] = -2.0
```

If you printed out `c`, you would get `[1.0, -2.0, 1.0]`. Why? You assigned two variables (`c` and `d`) to the *the same list*. So, when you use one reference (`d`) to change the list, you see the change if you look at the list from either reference. *FIXME: Diagram of two references to the same list here.*

To create two separate lists, you would need to explicitly make a copy:

```
1 c = [1.0, 2.0, 1.0]
2 d = c.copy()
3 d[1] = -2.0
```

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*This is a draft chapter from the Kontinua Project. Please see our website (<https://kontinua.org/>) for more details.*

# Answers to Exercises

